Contributions to the GDoT Data-Sufficiency Project

Kevin Li | April – August 2026 github.com/lzhkevin99-cell/gdot-data-sufficiency-contributions

Project Overview

When a Georgia DOT continuous count station (CCS) misses days of data, can its monthly average daily traffic still be trusted? The project's main paper answers this with a conformal-prediction procedure that estimates k^*, the minimum number of observed days needed. Alongside the paper, I carried out two independent workstreams that use its code and data read-only:

- **Exploratory data analysis** of the full 286-station network (2017–2024), run as an AI-agent pipeline that I designed.
- **A cluster-pooled reference-pool fallback**, after I found that the paper's pipeline silently drops 55% of eligible stations: a self-directed learning experiment, validated against baselines and not adopted.

My role

AI coding agents executed the code and analysis from specifications I wrote. I contributed the problem framing, the design of the pipeline and experiments, the specifications, and the evaluation of every result. Each number here was checked against the output that produced it.

1 Exploratory Data Analysis of the CCS Network

Objective

Establish what data exists, which parts are reliable, and how stations behave and relate, before any modeling.

Pipeline design and methodology

- Divided the analysis among **seven analysis agents and one synthesis agent**, each owning one question type, under a 475-line specification I wrote (`eda/claude.md`).
- Ordered the agents by data dependency. The quality audit runs before any behavioral analysis, so untrustworthy station-months are never treated as signal:

  Inventory → Quality audit → [Profiles · Trends · Corridors · Anomalies, in parallel] → Features → Synthesis

- Built two guardrails into the specification: a mandatory Risks/Caveats section for every agent, and explicit rules against overclaiming (e.g. no MCAR/MAR labels without evidence).

Key findings

Inventory	**286** stations with data out of a 330-station catalog, across three data tiers (5-min 2024, 5-min 2022–23, hourly 2017–22); 225 (78.7%) appear in all three.
Missingness	**21,837** station-months audited, **91.6%** complete ($\geq$80% bin coverage). Peaks in 2017 (13.6%) and 2022–23 ($\approx$12%, a data-pipeline transition) show it is operationally driven, not random.
Outage styles	A station-level classification not in the main paper: always-on 73.4%, occasional-gap 15.7%, seasonal-dropout 9.1%, chronic-sparse 1.8%.
Trust	**243 of 286** stations (84.9%) rated HIGH-trust, and 19,963 station-months (96.9%) trusted. Of 391 anomaly-days, 90.5% are explained by logged events.
Corridors	2,848 valid station pairs (mean $r = 0.573$); **27** reference-station candidates pass all five screening criteria.
Modeling readiness	A 286×27 station feature table and an 818,645-record clean hourly 2024 dataset.

Limitations and deliverables

- *Limitations, reported by the pipeline itself rather than hidden:* holiday-effect estimates return NaN for all six holidays (a station-ID join mismatch), and k=4 archetype clustering labels 77 of 78 stations "Mixed" (too few features to discriminate).
- *Deliverables:* a 14-page final report, a 14-slide final presentation, a 19-page summary report, a pipeline-design document, and 36 figures.

2 Cluster-Pooled Reference-Pool Fallback

A self-directed learning exercise: not adopted, and never presented as a change to the paper.

Problem identified

The paper's multi-site pipeline requires a reference pool of at least 19 correlated companion-months per target station-month. Tracing the code, I found that for April 2024 only **84 of 185** eligible stations meet this bar. The other **101 (55%)** are dropped silently: the function returns `None`, with no record of which stations failed or why. They skew low-volume and rural, exactly the sites a DOT most needs an answer for.

Hypothesis and experimental design

Hypothesis: stations with a similar weekly traffic pattern can supplement a too-small reference pool.

- *Profiles:* a 7-value day-of-week vector divided by the station's mean, so clusters reflect pattern, not road size.
- *Clustering:* K-means over $K = 2$–12. Silhouette peaks at $K{=}3$ (0.406); Calinski–Harabasz and Davies–Bouldin favor $K{=}4$–6. I kept $K{=}3$ by the pre-stated rule and documented the disagreement rather than call it optimal.
- *Integration:* a 0.90 completeness gate leaves 194 stations (clusters of 126 / 12 / 56). The station's own pool is always tried first via the paper's unchanged code; an assertion blocks target-data leakage.
- *Validation:* a non-circular holdout. 20 stations with large pools ($\geq$38) supply ground-truth k^*; each is shrunk below 19 and re-solved by the fallback and two simpler baselines.

Results and baseline comparison

Coverage: improved — **84** $\rightarrow$ **95** of 185 eligible stations (45% $\rightarrow$ 51%); 11 stations rescued.

Accuracy: not improved — Higher k^* error than the simplest baseline (table below).

Method	*n*	MAE (days)	RMSE	Bias	Corr.
Cluster fallback (this work)	100	3.48	4.85	+0.68	0.735
Baseline: same functional class	96	**2.91**	**4.30**	−1.26	**0.818**
Baseline: pool all stations	108	3.18	4.89	+2.10	0.787

k^* error vs. ground truth, 20 holdout stations × 6 missingness mechanisms (*n* = resolved pairs). Best values in bold.

Final decision and interpretation

- *Not adopted.* The simplest baseline won. A coverage gain that cannot be shown to preserve accuracy is not yet a contribution; the bar is the same-functional-class baseline, not "better than nothing." Deferred work (other algorithms, sensitivity, stability) is tagged in the run report.
- *Shape normalization is load-bearing,* independent of adoption. Raw-volume clustering agrees with the shape-based one at only **ARI = 0.028**: it merely splits the 18 busiest stations from 176 others whose weekly shapes are nearly identical.

3 Research Directions (Documented, Not Yet Tested)

1. *Mechanism-conditional pooling:* select companions conditional on the missingness mechanism under test.
2. *Outage co-occurrence as a negative signal:* reject companions whose outages coincide with the target's.
3. *Impute-then-conform:* fill small gaps from behavioral neighbors before the sufficiency test.
4. *Outage style as a k^* stratifier:* join the EDA's outage classes to existing k^* results.
5. *Exchangeability check:* do reference months spanning the 2022–23 transition violate the paper's assumption?

4 What This Work Demonstrates

- *Baselines before claims:* the coverage gain looked like success until the baseline comparison.
- *Reading research code closely:* tracing control flow exposed an undocumented 55% station loss.
- *Guardrails for AI-assisted analysis:* required caveats made the outputs report their own failures.